Classical Field Theory — Exercise Sheet 5

Cones with forcing and interior tails

Hints and answers

Write $\Box = \partial_t^2 - \Delta$. Every cone argument below concerns real C^2 fields on an open neighbourhood of the closed cone $\{0 \le t \le R,\ |x-a| \le R-t\}$, with $R > 0$. No existence theorem for a nonlinear equation is assumed or proved here.

Exercise 5.1 (Source work on a shrinking cone). Suppose $(\Box + m^2)w = F$, with $m \ge 0$ and continuous F. Let $E_m(t)$ be the integral over $B_{R-t}(a)$ of $e_m = (w_t^2 + |\nabla w|^2 + m^2 w^2)/2$.

(a) Derive the exact shrinking-ball balance, including source work and all lateral flux terms. Prove, for $0 \le t < R$,

$$E_m(t) \le e^t \left[E_m(0) + \frac{1}{2} \int_0^t e^{-s} \|F(s)\|^2_{L^2(B_{R-s}(a))} \mathrm{d}s \right].$$

(b) If $F = 0$ on the cone and both initial data vanish on its base, prove that $w = 0$ on the closed cone, including when $m = 0$. Explain why this is a local statement even if total energy is infinite.

Hint. The local balance has source term $w_t F$. Retain the inward motion of the ball, estimate source work by two squares, and multiply the resulting differential inequality by e^{-t}.

Answer. $E_m' = \int_{B_{R-t}} w_t F - \frac{1}{2} \int_{\partial B_{R-t}} [(w_t - \partial_n w)^2 + |\nabla_{\tan} w|^2 + m^2 w^2]$. Thus $E_m' \le E_m + \|F\|_2^2/2$, giving the bound. Zero data and source give zero field by integrating its vanishing velocity along vertical segments.

Exercise 5.2 (Finite speed for a semilinear equation, conditional on solutions). Let $V \in C^2(\mathbb{R})$ and suppose two real C^2 fields ϕ_1, ϕ_2 solve $\Box\phi + V'(\phi) = 0$ on a neighbourhood of the same closed cone. Assume their displacements and velocities agree on $B_R(a)$ at time zero.

(a) Prove that the fields agree on the closed cone. Use the mass-one energy of their difference, allowing V'' to have either sign. Give an explicit finite coefficient in the energy inequality in terms of a bound for $|V''|$ on the field values occurring in the cone.

(b) Suppose instead that the fields are defined on an open slab $I \times \mathbb{R}^d$, with $0 \in I$, and their two initial data agree outside a closed set K. Prove agreement at every (t, x) with $\mathrm{dist}(x, K) > |t|$. What extra condition makes the zero field an available comparison solution?

Hint. Write $V'(\phi_1) - V'(\phi_2) = b(\phi_1 - \phi_2)$ by integrating V'' along the segment between the two field values. In the equation for the difference, put $+w$ on the left and $(1-b)w$ on the right.

Answer. If L bounds $|V''|$ on the compact field range, the difference's mass-one cone energy satisfies $E_1' \le (1+L)E_1$. Zero initial energy implies agreement. Compact perturbations stay in $\mathrm{dist}(x, K) \le |t|$; comparison with zero requires $V'(0) = 0$.

Exercise 5.3 (Huygens' principle and the dimension). Consider the massless wave with zero displacement and real $g \in C_c^\infty(\mathbb{R}^d)$ as initial velocity. The following classical representation formulas may be used as given facts for $t > 0$:

$$u_1(t,x) = \frac{1}{2}\int_{x-t}^{x+t} g(y)\mathrm{d}y, \qquad u_2(t,x) = \frac{1}{2\pi}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\mathrm{d}^2y,$$

$$u_3(t,x) = \frac{1}{4\pi t}\int_{|y-x|=t} g(y)\mathrm{d}S_y.$$

For the higher-dimensional classification one may use these general forms. Let α_d be the volume of the unit ball, $⨍$ the integral divided by the region's volume or area, and $D_t = t^{-1}\partial_t$. For odd $d \geq 3$ and even $d \geq 2$, respectively,

$$u_d = \frac{1}{1\cdot 3\cdots(d-2)} D_t^{(d-3)/2}\left[t^{d-2}⨍_{|y-x|=t} g(y)\mathrm{d}S_y\right],$$

$$u_d = \frac{1}{2\cdot 4\cdots d} D_t^{(d-2)/2}\left[\frac{1}{\alpha_d}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\mathrm{d}^dy\right].$$

These facts hold for the smooth compact data specified here; see Evans, *Partial Differential Equations*, 2nd ed., §2.4.1, formulas (31), (38), Theorems 2–3, pp. 73–77. Nonzero displacement contributes the time derivative of the corresponding velocity formula with g replaced by that displacement.

(a) For $d = 1, 2, 3$, let $g \geq 0$, $g \not\equiv 0$, with support inside $B_1(0)$. Decide whether $u_d(t,0)$ vanishes for $t > 1$. Explain the difference between finite speed and dependence only on data arbitrarily near the base sphere.

(b) Use the general formulas to classify all positive spatial dimensions. For even d, test data supported strictly inside the base sphere so that every time derivative acts on a smooth kernel. Explain why a statement that all even-dimensional kernels are positive would be too strong.

Hint. For the tail test the integration domain contains the fixed support of g. Repeated application of D_t to $(t^2-r^2)^{-1/2}$ has alternating sign and a nonzero coefficient. In odd dimensions a whole neighbourhood of the base sphere missing the data makes all the displayed derivatives zero.

Answer. For $t > 1$, $u_1(t,0)$ and $u_2(t,0)$ are positive, while $u_3(t,0) = 0$. Huygens' sphere property holds for odd $d \geq 3$, but fails for $d = 1$ and every even d. Higher even-dimensional interior kernels alternate in sign.